VMWARE Networking, Windows Server Install Prep, Home Lab Architecture

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 2 hours and 30 minutes.

VMWare Networking Deep Dive

NAT Networking: Reminder: NAT = Network Address Translation, The VM shares your host computer's internet connection. VM gets a private IP from VMware. The VM can access the Internet. Other devices on your home network cannot directly reach the VM. It is "hidden" behind the host.

Bridged Networking: VM becomes another device on the physical network. VM gets an IP from the router. Appears as a completely separate computer. Other devices can communicate directly with the VM.

Host-Only: No internet, only host and VM communication. VM can talk to the host, cannot access the internet, and cannot reach your home network.

Run commands on Kali VM: **ip addr, ip route, and ping google.com**

Network isolation is the practice of separating systems, devices, or networks from other networks so that communication is limited or completely blocked. The goal is to prevent unwanted access, malware spread, unauthorized communication, or accidental damage.

Home Lab Architecture Planning

Rough Structure:

```
Internet
  ↓
Host Machine
  ↓
VMware
  ├── Kali Linux
  ├── Windows Server
  ├── Windows Client
  └── Future Security Tools
```

A domain controller lab is a practice environment where you build and manage a Windows domain just like an enterprise would, usually using VMs. The purpose is to learn how enterprise authentication, user management, policies, and network services work without touching a real production environment.

A Windows Client VM is a virtual machine running a client OS such as Windows 11 or Windows 10.

Snapshots let you return to a state of the VM. Useful is something bad happens.

Windows Server Installation Prep

Obtain Windows Server ISO. Download one.

An ISO file is a single file that contains an exact copy of an entire CD, DVD, or installation disc. It is a digital version of a physical disc.

UEFI vs BIOS: When a computer powers on, it needs firmware that starts the hardware and loads the operating system. The two main firmware systems are: BIOS (older) and UEFI (modern replacement). BIOS stands for Basic Input/Output System.

Power On
↓
BIOS Starts
↓
Checks Hardware (POST)
↓
Looks for Boot Device
↓
Loads Operating System

UEFI stands for Unified Extensible Firmware Interface. It is the modern replacement for Bios.
Power On
↓
UEFI Starts
↓
Checks Hardware
↓
Loads Boot Manager
↓
Starts Windows